Problem1 Code:

Saniya_Bagwan.java

```
import java.util.HashMap;
import java.util.Scanner;

abstract class Student{
   private int studentId;
   private String name;
   private String course;
   private float baseFee;

   public Student(int studentId, String name, String course, float baseFee){
      this.studentId=studentId;
      this.name=name;
      this.course=course;
      this.baseFee=baseFee;
   };

   void display(){
      System.out.println("Student ID: "+studentId);
      System.out.println("Student Name: "+name);
      System.out.println("Student Course: "+course);
      System.out.println("Student BaseFee: "+baseFee);
   }

   public String getName() {
      return name;
   }
   public void setName(String name) {
      this.name = name;
   }
   public float getBaseFee() {
      return baseFee;
   }
   public void setBaseFee(float baseFee) {
      this.baseFee = baseFee;
   }

   public int getStudentId() {
      return studentId;
```

```
    }

    public void setStudentId(int studentId) {
        this.studentId = studentId;
    }

    public String getCourse() {
        return course;
    }

    public void setCourse(String course) {
        this.course = course;
    }

    public abstract double calculateFee();
}

class RegularStudent extends Student{

    public RegularStudent(int studentId, String name, String course, float baseFee){
        super(studentId,name,course,baseFee);
    }
    public double calculateFee(){
        return getBaseFee();
    }
}

class ScholarshipStudent extends Student{
    public ScholarshipStudent(int studentId, String name, String course, float baseFee){
        super(studentId,name,course,baseFee);
    }
    public double calculateFee(){
        return getBaseFee() -0.2*getBaseFee();
    }
}

class ExchangeStudent extends Student{
    private static final float international_charges=12000;
    public ExchangeStudent(int studentId, String name, String course, float baseFee){
        super(studentId,name,course,baseFee);
    }
```

```
    public double calculateFee(){
        return getBaseFee() +international_charges;
    }
}

public class Saniya_Bagwan {
    public static void main(String[] args) {
        HashMap<Integer,Student> studentMap= new HashMap<>();
        int c,id,t;
        String n,cs;
        float b;
        Scanner sc = new Scanner(System.in);
        do {
            System.out.println();
            System.out.println("-------------Menu-------------");
            System.out.println("1.Add Student");
            System.out.println("2.Search Student");
            System.out.println("3.Delete Student");
            System.out.println("4.Update Student");
            System.out.println("5.Display All Students(Polymorphism )");
            System.out.println("6.Exit");
            System.out.println();
            System.out.print("Enter Choice: ");
            c = sc.nextInt();
            switch (c) {
                case 1:
                    System.out.print("Enter Student ID: ");
                    id = sc.nextInt();
                    sc.nextLine();
                    System.out.print("Enter Student Name: ");
                    n = sc.nextLine();
                    System.out.print("Enter Student Course: ");
                    cs = sc.nextLine();
                    System.out.print("Enter Student Basefee: ");
                    b = sc.nextFloat();
                    sc.nextLine();
                    System.out.println();
                    do {
                        System.out.println("Select Student Type: ");
                        System.out.println("1.Regular");
                        System.out.println("2.Scholarship");
```

```
        System.out.println("3.Exchange");
        System.out.println();
        System.out.print("Enter Choice: ");
        t = sc.nextInt();
        switch (t) {
            case 1:
                studentMap.put(id, new RegularStudent(id, n, cs, b));
                break;
            case 2:
                studentMap.put(id, new ScholarshipStudent(id, n, cs, b));
                break;
            case 3:
                studentMap.put(id, new ExchangeStudent(id, n, cs, b));
                break;
            default:
                System.out.println("Invalid Type Selected!");
        }
    } while (t != 1 && t != 2 && t != 3);
    System.out.println("Student Added!");
    break;
case 2:
    System.out.print("Enter Student ID: ");
    id = sc.nextInt();
    sc.nextLine();
    if (studentMap.containsKey(id)) {
        System.out.println("Student Found!");
        studentMap.get(id).display();
    } else {
        System.out.println("Student Not Found!");
    }
    break;
case 3:
    System.out.print("Enter Student ID: ");
    id = sc.nextInt();
    sc.nextLine();
    if (studentMap.containsKey(id)) {
        System.out.println("Student Found!");
        studentMap.remove(id);
        System.out.println("Student with ID :" + id + " is Deleted!");
    } else {
        System.out.println("Student Not Found!");
```

```
    }
    break;
  case 4:
    System.out.print("Enter Student ID: ");
    id = sc.nextInt();
    sc.nextLine();
    if (studentMap.containsKey(id)) {
      System.out.println("Student Found!");
      System.out.println();
      System.out.println("Old Details: ");
      studentMap.get(id).display();
      System.out.println();
      System.out.print("Enter Student Name: ");
      n = sc.nextLine();
      System.out.print("Enter Student Course: ");
      cs = sc.nextLine();
      System.out.print("Enter Student Basefee: ");
      b = sc.nextFloat();
      sc.nextLine();
      System.out.println();
      do {
        System.out.println("Select Student Type: ");
        System.out.println("1.Regular");
        System.out.println("2.Scholarship");
        System.out.println("3.Exchange");
        t = sc.nextInt();
        switch (t) {
          case 1:
            studentMap.put(id, new RegularStudent(id, n, cs, b));
            break;
          case 2:
            studentMap.put(id, new ScholarshipStudent(id, n, cs, b));
            break;
          case 3:
            studentMap.put(id, new ExchangeStudent(id, n, cs, b));
            break;
          default:
            System.out.println("Invalid Type Selected!");
        }
      } while (t != 1 && t != 2 && t != 3);
      System.out.println("Student Data Updated!");
```

```
            } else {
                System.out.println("Student Not Found!");
            }
            break;
        case 5:
            if(studentMap.size()==0){
                System.out.println("No Data Found!");
            }else {
                System.out.println("All Student Fees: ");
                for (Student s : studentMap.values()) {
                    s.display();
                    System.out.println("Calculated Fees: " + s.calculateFee());

                }
            }
            break;
        case 6:
            System.out.println("Exiting.........");
            break;
        default:
            System.out.println("Invalid Choice!");
        }
    }while (c!=6);
    sc.close();
  }
}
```

Output:

Inserting All type of Students

```
Run  Saniya
"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.3\lib\idea_rt.jar=56379" -Dfile.encoding=UTF-8 -Dsun.stdout.encodin

-------------Menu-------------
1.Add Student
2.Search Student
3.Delete Student
4.Update Student
5.Display All Students(Polymorphism )
6.Exit

Enter Choice: 1
Enter Student ID: 1
Enter Student Name: Saniya Bagwan
Enter Student Course: IT
Enter Student Basefee: 100000

Select Student Type:
1.Regular
2.Scholarship
3.Exchange

Enter Choice: 2
Student Added!

-------------Menu-------------
1.Add Student
2.Search Student
3.Delete Student
4.Update Student
```

```
-------------Menu-------------
1.Add Student
2.Search Student
3.Delete Student
4.Update Student
5.Display All Students(Polymorphism )
6.Exit

Enter Choice: 1
Enter Student ID: 2
Enter Student Name: Payal Lokhand
Enter Student Course: Civil Engineering
Enter Student Basefee: 100000

Select Student Type:
1.Regular
2.Scholarship
3.Exchange

Enter Choice: 1
Student Added!
```

```
-------------Menu-------------
1.Add Student
2.Search Student
3.Delete Student
4.Update Student
5.Display All Students(Polymorphism )
6.Exit

Enter Choice: 1
Enter Student ID: 3
Enter Student Name: John Abrahm
Enter Student Course: AI
Enter Student Basefee: 100000

Select Student Type:
1.Regular
2.Scholarship
3.Exchange

Enter Choice: 3
Student Added!
```

Searching :

```
-------------Menu-------------
1.Add Student
2.Search Student
3.Delete Student
4.Update Student
5.Display All Students(Polymorphism )
6.Exit

Enter Choice: 2
Enter Student ID: 2
Student Found!
Student ID: 2
Student Name: Payal Lokhand
Student Course: Civil Engineering
Student BaseFee: 100000.0
```

Updating:

```
-------------Menu-------------
1.Add Student
2.Search Student
3.Delete Student
4.Update Student
5.Display All Students(Polymorphism )
6.Exit

Enter Choice: 4
Enter Student ID: 3
Student Found!

Old Details:
Student ID: 3
Student Name: John Abrahm
Student Course: AI
Student BaseFee: 100000.0

Enter Student Name: John Abrahm
Enter Student Course: AI & DS
Enter Student Basefee: 100000

Select Student Type:
1.Regular
2.Scholarship
3.Exchange
3
Student Data Updated!
```

Display For Demonstrating Polymorphism:

```
-------------Menu-------------
1.Add Student
2.Search Student
3.Delete Student
4.Update Student
5.Display All Students(Polymorphism )
6.Exit

Enter Choice: 5
All Student Fees:
Student ID: 1
Student Name: Saniya Bagwan
Student Course: IT
Student BaseFee: 100000.0
Calculated Fees: 80000.0
Student ID: 2
Student Name: Payal Lokhand
Student Course: Civil Engineering
Student BaseFee: 100000.0
Calculated Fees: 100000.0
Student ID: 3
Student Name: John Abrahm
Student Course: AI & DS
Student BaseFee: 100000.0
Calculated Fees: 112000.0
```

Delete:

```
-------------Menu-------------
1.Add Student
2.Search Student
3.Delete Student
4.Update Student
5.Display All Students(Polymorphism )
6.Exit

Enter Choice: 3
Enter Student ID: 3
Student Found!
Student with ID :3 is Deleted!
```

Display:

```
-------------Menu-------------
1.Add Student
2.Search Student
3.Delete Student
4.Update Student
5.Display All Students(Polymorphism )
6.Exit

Enter Choice: 5
All Student Fees:
Student ID: 1
Student Name: Saniya Bagwan
Student Course: IT
Student BaseFee: 100000.0
Calculated Fees: 80000.0
Student ID: 2
Student Name: Payal Lokhand
Student Course: Civil Engineering
Student BaseFee: 100000.0
Calculated Fees: 100000.0
```